

When does a geodesic probability path help latent flow matching? A controlled study on a two dimensional Navier-Stokes surrogate

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1 Motivation

Neural PDE surrogates are usually trained to regress the next state from the current one. A generative version can be more useful, because sampling the next state from a conditional distribution lets the model show a spread of outcomes instead of one blurred mean. Flow matching [4] fits this, since it trains a generative model by regression on a velocity field and needs no sampling during training.

Flow matching has one part that is rarely questioned: the probability path, the family of distributions that links a prior to the data as a synthetic time τ runs from zero to one. The usual path is a straight line between a noise sample and a data sample. The surrogate here works not on raw vorticity fields but on compressed latent tokens, which are ordinary vectors, so how to interpolate between them is a real choice. The question of this report is simple: does that choice change the result for a latent PDE surrogate, and if so, when.

1.1 The framework

Riemannian flow matching [1] does not assume the latent space is a sphere. It assumes the data lie on some manifold M and replaces the straight line with the geodesic between the prior and data samples. The assumption is $x \in M$ for a suitable M , not $M = S^{D-1}$. The sphere is one member of a family that also includes tori, rotation groups, and learned manifolds, and the same construction gives back the straight line as its flat case. The experiment here picks the unit sphere S^{31} in the thirty two dimensional token space, so it tests one manifold, not the framework.

1.2 Hypotheses

The sphere is a reasonable first choice: it is the simplest manifold with closed form exponential and logarithm maps, and a common idea in representation learning is that a code's direction carries its meaning while its length is a nuisance, which would put codes on a sphere. That idea can be tested. The starting hypothesis (H1) is that a geodesic path is a better inductive bias than a straight line, so the sphere arm should win on accuracy, on few step accuracy, or on sample quality, with everything else fixed. The null (H0) is that the two paths give the same result once the network has enough capacity. The ablation in Section 5 leads to a sharper hypothesis (H2): a geodesic path helps only when the data actually lie on the assumed manifold.

2 Background

This section sets the terms the results and figures use. The equations are in Appendix C.

2.1 The two stage pipeline

Stage one trains a vector quantised autoencoder [10] that compresses each 64×64 vorticity field into an 8×8 grid of thirty two dimensional tokens from a learned codebook. Stage two freezes that latent space and trains the flow matching model to generate the next frame’s tokens from the current frame’s. Freezing stage one keeps the comparison clean: both arms share the same latent space, so any difference between them comes from the path and not the autoencoder.

2.2 Two facts about the paths

The straight line path has a velocity that stays constant along the path, so a single Euler step from the prior already lands near the target, and a straight field is cheap to integrate. The sphere path replaces the line with the shortest arc between two directions. In high dimension two random directions are almost at right angles: their inner product has mean zero and variance $1/D$, so with $D = 32$ the angle sits near ninety degrees (Figure 1). Each conditional geodesic is then a long quarter turn arc, and the field the network has to learn is more curved and harder to integrate in few steps. Straight is cheap to integrate and curved is not. These two facts are enough to read the plots below.

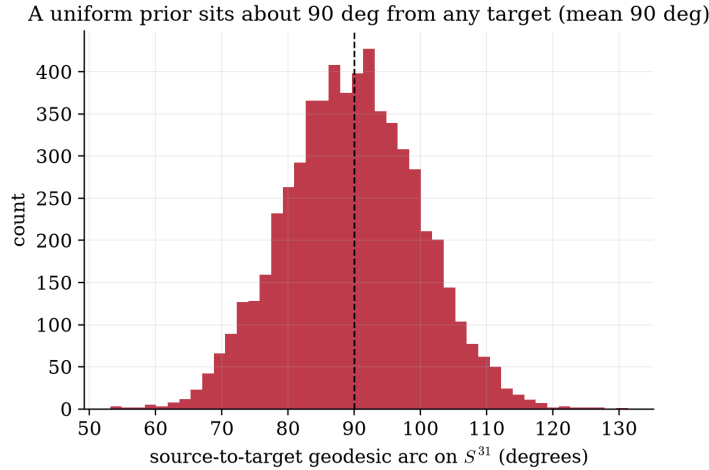


Figure 1: Angle between a random prior direction and an arbitrary target on S^{31} , over twenty thousand pairs. It sits near ninety degrees, so a uniform prior places almost every source a quarter turn from its target. This is why the geodesic field is more curved than the straight line field.

2.3 The magnitude head

This part matters most for reading the result. A token splits into a direction and a length, $x = r \hat{x}$ with $r = \|x\|$. The straight line arm models the whole vector. The sphere arm lives on the unit sphere, where every point has length one, so it can only model the direction and has no way to represent a length. When it generates a direction for the next frame, the length has to come from somewhere. The simplest option is to reuse the current frame’s length. The magnitude head does a little better: for each token it predicts the log ratio $\log(r_{\text{next}}/r_t)$ from the current state, trained by regression next to the flow. It is the one difference between the arms, and it patches a gap the sphere creates rather than being a setting of the path. The number to keep in mind is how little it has to do: on this data a token’s length changes by about two percent per frame, so the head is

close to doing nothing, and Section 5.3 shows this is why the sphere’s missing length costs almost nothing here.

2.4 Metrics

- **Rollout relative L_2 :** run the model forward through the whole trajectory, feeding its own output back in with no teacher forcing, then take the relative L_2 error of the final frame against the truth. This is the standard operator learning error [3], and it is harder than one step prediction because errors build up over the rollout.
- **Final frame MSE:** the pointwise squared error on the last frame.
- **Spectral error:** the relative error between the radially averaged energy spectra of the generated and true fields. It checks whether energy sits at the right spatial scales, which the pointwise error can miss.
- **NFE:** the number of integration steps per generated step, the accuracy against cost axis.
- **One step penalty:** how much worse a single step is than an arm’s own best step count, a short measure of how step robust a path is.
- **Energy distance and MMD:** two sample distance measures used in the ablation. The energy distance [9] has no bandwidth to set, so it cannot be tuned to favour one arm.
- **Radius coefficient of variation and intrinsic dimension:** checks of where the latents sit. A radius coefficient of variation near zero means a thin shell, and the TwoNN intrinsic dimension [2] estimates how many dimensions the data use locally.

3 Experiment setup

3.1 Controlled design

Stage one is trained once and frozen. Stage two trains two flow matching models on that frozen space with the same network, optimiser, batch size, learning rate, and seed, changing only the path. Any difference between the arms then comes from the path geometry. This is also what lets a null result mean something: if two arms that share everything but the path come out equal, the path did not change the outcome, rather than one arm being tuned worse.

3.2 Autoencoder, velocity network, and data

The autoencoder is a transformer with an exponential moving average quantiser. On top of the reconstruction and commitment terms, its loss adds a physics residual on the decoded field, with velocity recovered from vorticity through the streamfunction, spatial derivatives taken by spectral differentiation, and the residual scaled by the root mean square time derivative so a single physics weight works across settings. Ramping that weight over the first epochs avoids an early instability from the quadratic advection term while the decoder is still poor. Figure 2 shows stage one training, and Figure 3 shows that the frozen latent space reconstructs held out fields well, which the later comparison needs.

The velocity network reads the interpolated tokens, the flow time through adaptive layer normalisation [8], and the current state as extra tokens; it is the same for both arms. The straight line arm regresses the difference between data and prior and steps from a Gaussian prior. The sphere

arm regresses the tangent velocity of the arc, projects onto the tangent space, and steps with the exponential map from a uniform prior on the sphere, so every point stays on the sphere; its one extra part is the magnitude head. The data are two dimensional Navier-Stokes vorticity in the Fourier neural operator setup [3]. Shapes are in Appendix A.

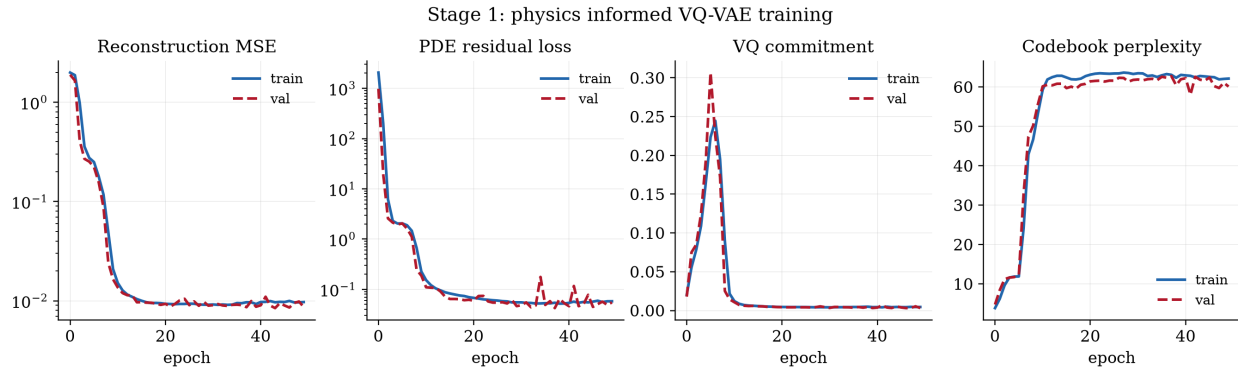


Figure 2: Stage one training on log axes. The physics residual starts about three orders of magnitude above its steady value, the quadratic advection effect, then settles. Perplexity rises to about sixty active codes, so the codebook does not collapse.

3.3 Seeds and evaluation

The results that carry weight come from three training seeds run to convergence, because the single seed and three seed pictures disagree on every metric that looked clear at one seed. Both arms in a seed share that seed. Evaluation is autoregressive rollout with the sampling reseeded before each arm and each step count, so the numbers are repeatable and both arms see the same random draws.

VQ-VAE reconstruction of held-out vorticity fields

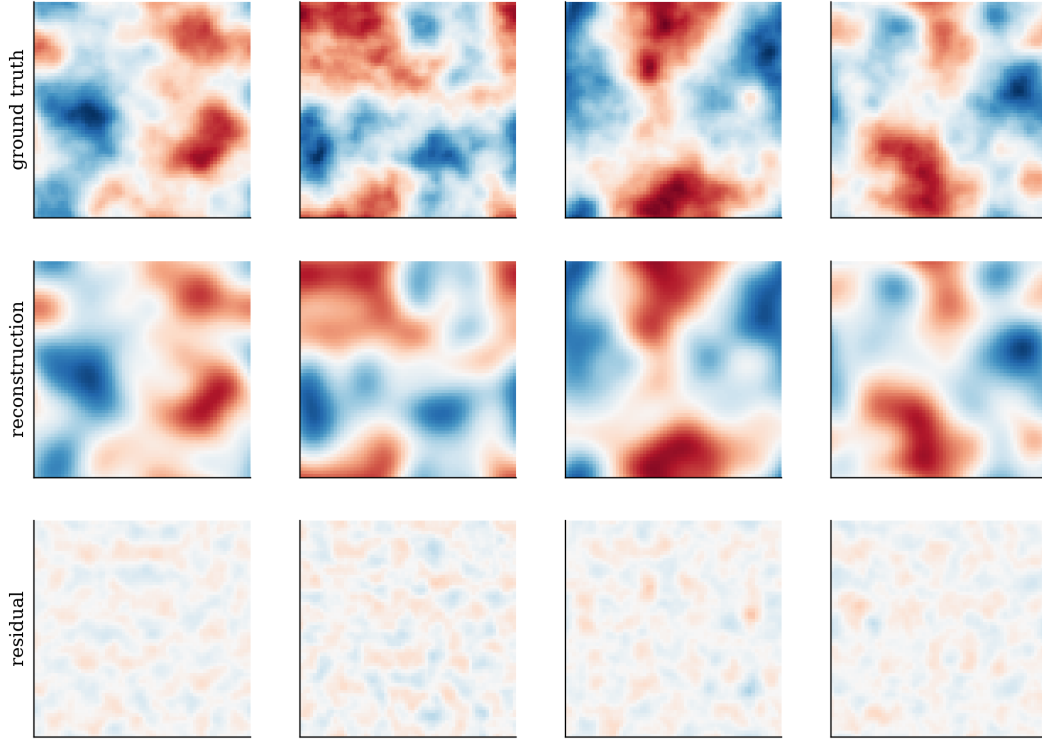


Figure 3: Reconstruction of held out fields. Top row ground truth, middle row reconstruction through the frozen autoencoder, bottom row residual on the same colour scale. The latent space keeps the main structures, so the comparison is not limited by reconstruction error.

4 Results

The comparison runs three training seeds, both arms, two hundred epochs each. Training reached convergence: the stage two loss is flat over the last fifty epochs for every arm and seed. A single seed run at seed forty two is shown only as an illustration, since its details did not repeat across seeds.

4.1 The two paths come out equal

Table 1 and Figure 4 hold the main result. Over three converged seeds, neither path is reliably ahead. On rollout relative L_2 and final frame MSE the sphere arm wins two of the three seeds and has the lower mean, but the seed to seed spread is wider than the gap between the means, so this is a small lean and not a win. On spectral error the ranking is just seed noise: seed zero favours the straight line by about nineteen times, seed one favours the sphere by about fourteen times, and seed two is close. There is no stable ranking to report on the spectrum.

Table 1: Three converged seeds at the reference step budget. Mean $\pm$ standard deviation over seeds zero, one, two; lower is better everywhere. The first single seed run had shown a clear straight line win on spectral error, which three seeds show to be noise.

Metric	Euclidean	Spherical	Seeds won
Rollout rel. L_2	0.0766 ± 0.0032	0.0711 ± 0.0071	Spherical 2/3
Final frame MSE ($\times 10^{-4}$)	1.22 ± 0.11	1.07 ± 0.24	Spherical 2/3
Spectral error	0.014 ± 0.012	0.031 ± 0.035	no stable winner

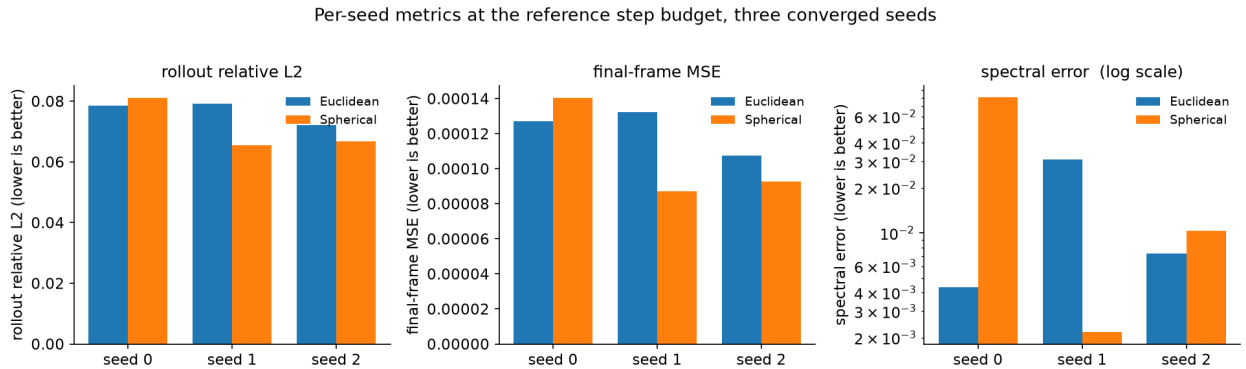


Figure 4: Per seed metrics at the reference step budget, three converged seeds. Read the accuracy panels (left, middle) as a small lean toward the sphere that the bars overlap on. The spectral panel (right, log scale) is the point to notice: the ranking swaps by more than ten times in both directions across seeds, so the single seed spectral win for the straight line is not real.

4.2 Step budget robustness leans one way but is not a rule

The straight line’s constant field should integrate in one step and the geodesic arc should not, so the straight line should be more step robust. Figure 5 shows this holds on two of three seeds and reverses on the third. On seeds one and two the straight line is flat and near its best at one step, while the sphere pays a fourteen to nineteen percent one step penalty. On seed zero the pattern

flips: the straight line has a thirty nine percent one step penalty and the sphere is at its best at one step. Averaged over seeds the one step penalties are about the same, thirteen percent for the straight line and eleven for the sphere. So the expected pattern is a tendency, not a rule, probably because nothing forces the trained straight line field to actually stay straight, and on seed zero it did not.

One effect does hold on every seed and both arms: accuracy is best at one to three steps and gets worse toward fifty. A correct solver on a correct field would not get worse with more steps, so this probably means the learned field, not the integrator, sets the accuracy here. It also means the common reference of eight steps already sits past the best point, so more integration steps do not help on this task.

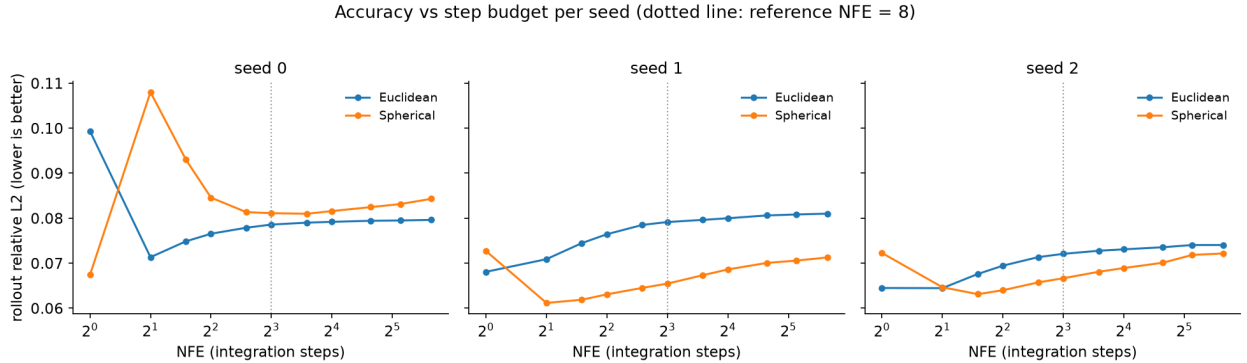


Figure 5: Accuracy against step budget, one panel per seed. On seeds one and two the straight line is flat and the sphere dips then rises (its one step penalty). On seed zero the two swap. On all three, both arms are best at one to three steps and worse beyond, so the marked reference of eight steps is already past the best point.

4.3 What the controls did, and the generated fields

The path of the numbers is worth stating, because it shows why the tie is trusted more than the first result. The first version, unseeded and single seed, showed the straight line ahead on almost every metric, including a spectral error about five times better. Each control that was added moved the result toward a tie: seeding the two arms the same removed a lead the straight line got from a luckier draw, training to convergence let the slower sphere arm catch up on rollout, and going to three seeds showed the spectral and step robustness wins were seed dependent. A result that weakens every time a control is tightened is not one to rely on. Figure 6 shows both arms tracking the truth by eye, which fits the tie. The raw stage two training losses are not plotted because they are not comparable across arms: the straight line regresses a difference whose scale is set by the Gaussian prior, while the sphere regresses a bounded tangent vector, so the sphere’s loss is smaller by construction and says nothing about which model is better.

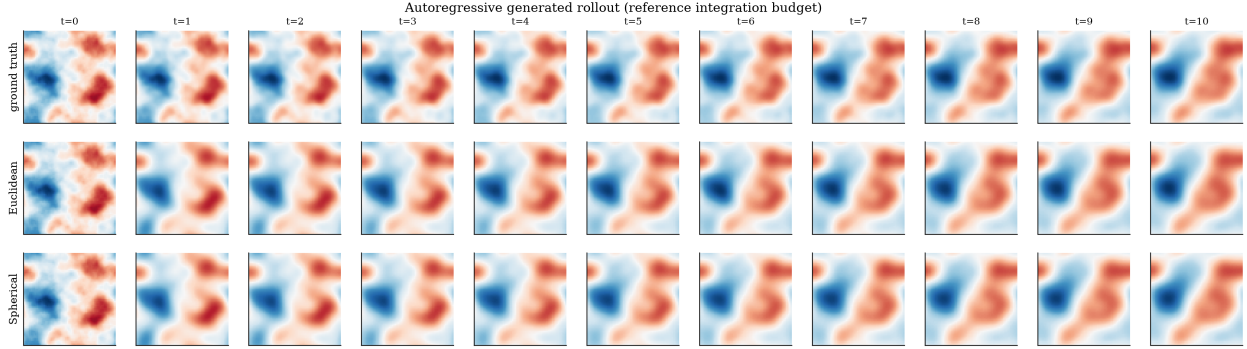


Figure 6: Generated vorticity fields over the rollout horizon at the reference budget, one seed. Top row ground truth, middle row the straight line path, bottom row the sphere. Both give coherent fields, which fits the equal scores.

5 Ablation studies

A tie on its own is weak, because it can also mean the measurement was not careful enough. The ablation turns the tie into an understood one: it isolates the variable that would make geometry matter, shows it matters when present, measures that it is absent from the real latents, and then looks at why its absence gives a tie here rather than a penalty.

5.1 The controlled ablation

The test needs two datasets that are the same except in manifold structure. Because the flow matching loss is minimised by a known conditional expectation (Appendix C), a tiny multilayer perceptron can read the path comparison at its target rather than through a heavy network; its hand written gradients match finite differences to a relative error of 10^{-7} . The two regimes share the same directional distribution and differ only in whether every point has unit length (on manifold) or a length that depends on direction (off manifold). Figure 7 reports the geodesic penalty, the ratio of the geodesic to the straight line energy distance, where above one means the sphere is worse.

Over three seeds the on manifold penalty averages 0.63 and the off manifold penalty 2.17, and the straight line wins all three off manifold seeds. The part that holds across seeds is the one sided pattern: off the manifold the geodesic is clearly worse, and on the manifold it is about equal or a little better. This is the flip H2 predicts. It also matches the source material, where the only Riemannian example in the reference library [5] is a flat torus, which is data that really lies on a manifold, while the ordinary vector data uses the straight line.

5.2 The real latents are off the sphere

The sphere idea predicts that every token has the same length and that the length carries no information beyond direction. Both predictions fail (Figure 8): the token length has a coefficient of variation near 0.27, and it correlates with direction at about 0.21, which puts the latents on the off manifold side of the ablation. The intrinsic dimension is about five and a half, well below the ambient thirty two, so the latents sit near a low dimensional manifold that is not the sphere. This rules out the sphere; it does not identify the true manifold.

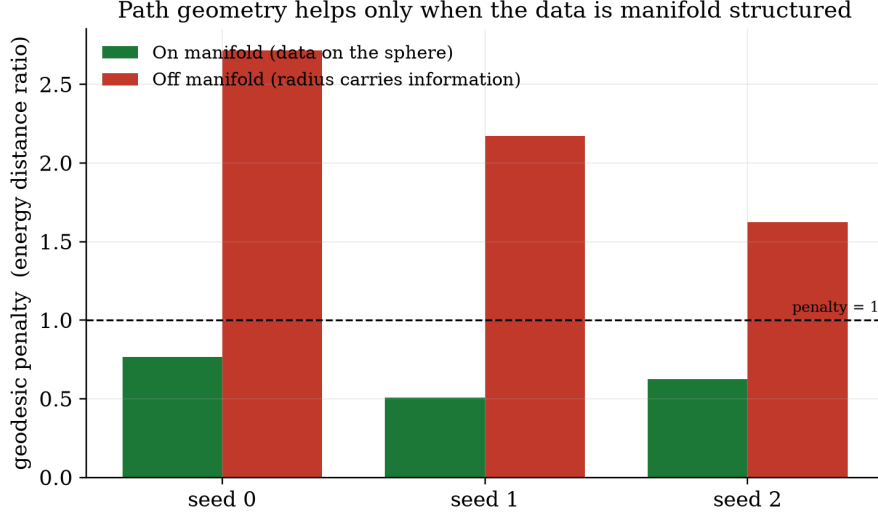


Figure 7: The controlled ablation, three seeds. The directional distribution is the same in both regimes; only the manifold structure changes. On the manifold the geodesic is better in every seed; off the manifold it is worse in every seed, by about one and a half to three times.

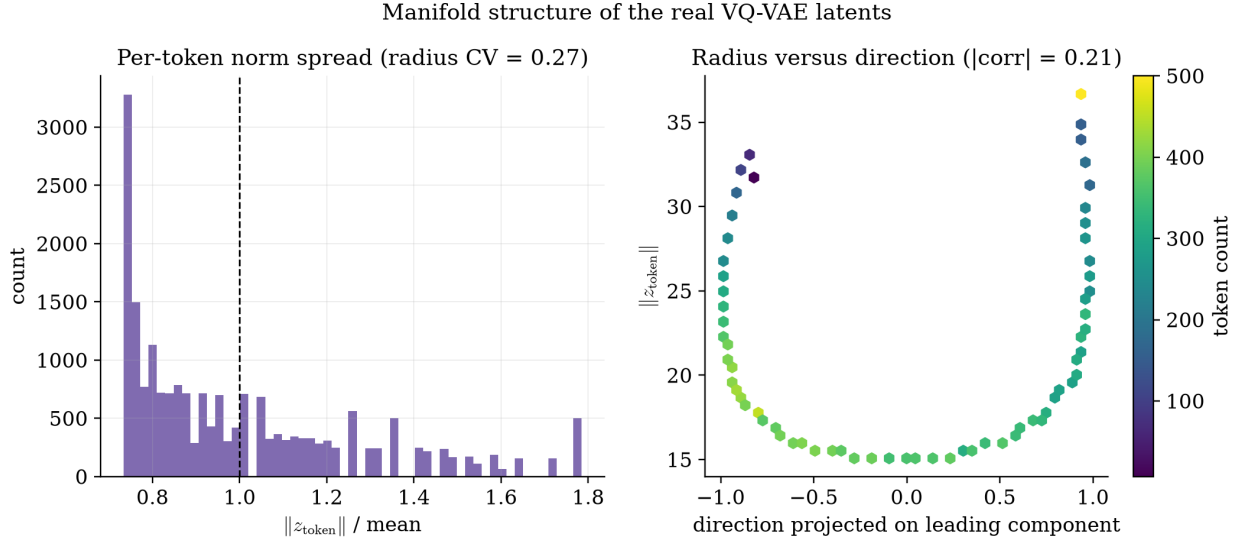


Figure 8: Geometry of the real latents. Left, the per token length has a coefficient of variation near 0.27, so the latents are not a thin shell. Right, the length correlates with direction, so the sphere’s split into direction and separate length drops a real link between them. The latents are off manifold, the case where the ablation predicts a penalty.

5.3 Why the ablation predicts a penalty that the real task does not show

The ablation predicts a penalty off the manifold, the latents are off the manifold, and yet the converged task shows no penalty. The likely reason, which the numbers support but this study does not fully isolate, is the difference between the two settings. The ablation is unconditional: its model builds a sample from noise, so it has to produce both a direction and a length, and the sphere drops the length and has to invent it, which is costly when length depends on direction. The same unconditional test on the real encoder latents repeats the flip, with the straight line winning every seed on the raw latents (energy distance 0.21 against 0.41) and the sphere winning every seed once the latents are placed on the sphere (0.008 against 0.043). The real task is not unconditional. It is a next step problem, and the length the sphere drops is carried from the current frame, where it changes by only about two percent per step (Section 2.3). So the sphere probably keeps almost the right length for free and its flow only has to model the direction. On this reading the part of the penalty that the ablation measures is nearly switched off here. Figure 9 separates the two effects: the spatial length spread that the split drops is large, while the frame to frame length change that the head has to supply is small, and the current frame supplies it.

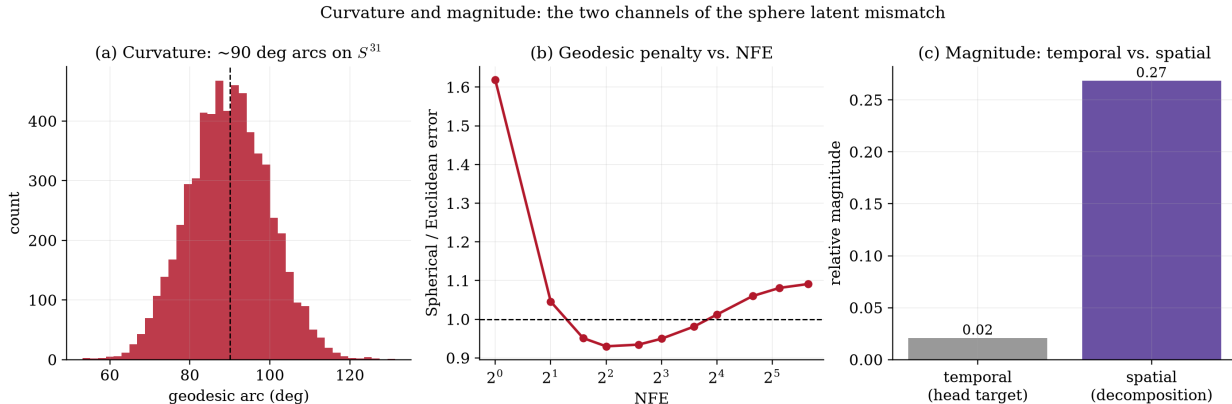


Figure 9: The two effects the mismatch could act through. (a) Curvature: source to target arcs on S^{31} sit near ninety degrees. (b) A single run trace, the geodesic to straight line error ratio; the three seed version (Figure 5) shows it is only a tendency. (c) The length effect: the frame to frame change the head supplies is small, while the spatial spread the split drops is large, so the current frame nearly covers what the sphere loses.

5.4 The magnitude head does very little, and reproducibility

The head does very little, which fits the reading above. At the start its loss equals the mean square of the true log length ratio, the two percent per step change, and freezing, predicting, or dropping it moves the result by a fraction of a percent. Two reproducibility details set how far the numbers can be trusted. Seeding the two arms the same removed a lead the straight line got from a luckier start, and is much of why the converged sphere arm is even. Atomic checkpoint writes matter because the three seed run is long enough to be interrupted, and a half written checkpoint would be picked up as the newest file and fail to load.

6 Discussion

The motivating question was whether the path geometry changes the result and, if so, when. On this setup the answer to the first part is that it did not: over three converged seeds the straight line and the sphere geodesic score the same on accuracy, the same on the spectrum once seed noise is taken out, and about the same on step robustness. The ablation gives a likely answer to the second part. A geodesic path seems to help only when the data sit on the assumed manifold, and these latents do not, so the sphere gives no benefit here. It also seems to give no penalty, probably because the task is conditional and the one thing the sphere drops, the length, is nearly supplied by the current frame. Both of these are supported by the measurements but not proven, so they are best read as the most likely explanation rather than a settled fact.

It is worth being clear about the size of the claim. This is a null result on one manifold, one encoder, and one small task, at three seeds. It shows an absence of a measurable difference, not proof that the two paths are identical. What does seem to carry over is narrower and more useful: geometric structure has to be earned from the measured representation before it helps, and on a conditional next step task even a real mismatch may stay hidden in the headline metrics because the conditioning covers for it.

7 Conclusion

Path geometry was expected to help a latent Navier-Stokes surrogate, and on this setup it did not change the result. Over three converged seeds with the arms controlled down to the seed, the straight line and the sphere geodesic score the same on accuracy and on the spectrum once seed noise is removed, and about the same on step robustness. The clear straight line win from the first single seed run did not survive seeding, convergence, and repeats. The ablation gives a consistent account of the tie: the sphere does not match the off manifold latents, so it gives no benefit, and the conditional task probably supplies the length the sphere drops, so it gives no clear penalty. A fair reading is that path geometry is not a useful lever on a conditional latent surrogate whose representation is not built to match it, and that the straight line is a sensible default here.

8 Limitations and future work

Limitations. Three seeds is enough to overturn a single seed headline but is not many, and the small accuracy lean toward the sphere sits inside the seed spread, so it is not claimed as real. Only one manifold was tested, the sphere, out of the many the framework allows. The latent geometry is a property of the encoder, not of the equation, so a different encoder could move where the latents sit. And the main explanation in Section 5.3, that the conditional task covers for the sphere’s dropped length, is supported by the two percent per step change and the near zero magnitude head, but it is not isolated by its own experiment, so it stays a likely explanation rather than a proven one.

Future work. Each gap above suggests a test. The seed count can be raised to five or ten with error bars to check whether the small accuracy lean is real or noise. Other manifolds from the same family, a torus or a learned manifold, can be tried in place of the sphere. The encoder dependence can be checked by repeating the study on a different encoder. The reconciliation can be tested directly by making the sphere arm generate its length from noise rather than carrying it from the current frame, and by moving to a task with a larger frame to frame length change: if the explanation is right, the ablation penalty should then appear on the real task.

References

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A Model architectures

The velocity network is the same for both arms. The magnitude head is only in the sphere arm and is the one recorded difference.

Table 2: Architecture summary. The velocity network is shared by both arms.

Component	Setting	Value
VQ-VAE	input resolution	64×64
	patch size	8
	latent grid	8×8 tokens
	latent dimension D	32
	codebook size	512 (EMA updates)
	transformer width	128
	depth / heads	6 / 8
Velocity net	working width	128
	depth / heads	6 / 8
	flow time conditioning	adaptive layer norm [8]
	state conditioning	in context tokens with type embedding
	flow time embedding dim	128
Magnitude head (spherical only)	width	64
	depth / heads	4 / 8
	target	per token $\log(r_{\text{next}}/r_t)$

B Hyperparameters, sweeps, and reproducibility

Data. Two dimensional Navier-Stokes vorticity, resolution 64, viscosity $\nu = 10^{-3}$, save interval 0.1, Gaussian random field initial condition following Li et al. [3]. Evaluation uses sixteen held out test trajectories of twenty frames each.

Optimisation. Both stages use AdamW [7] with learning rate 2×10^{-4} , weight decay 10^{-4} , batch size 32. Stage one runs fifty epochs with a five epoch linear ramp on the physics weight. Stage two runs two hundred epochs per arm, to convergence.

Seeds and sweeps. The main results are three training seeds, 0, 1, 2, each training both arms from scratch on the same frozen autoencoder, followed by the seeded evaluation and the step budget sweep. A single seed run at seed 42 is shown only as an illustration. The step budget sweep covers 1, 2, 3, 4, 6, 8, 12, 16, 25, 35, 50 steps, denser at the low end, and reports the final rollout horizon error at the reference budget of eight. Sampling is reseeded before each arm and each step count. The multi seed training and the sweep are each one resumable script.

Software. The controlled ablation is self contained and needs neither a GPU nor a checkpoint. Separate scripts reproduce the geometric checks, the unconditional real latent test, and, from the saved multi seed CSVs, both multi seed figures. Code and configurations come with this report.¹ Data generation is a CPU bound spectral solver; the two training stages target one modest GPU.

¹Project repository: <https://github.com/AmiteshPuri/FM-paths>

C Mathematical details

Conditional flow matching. Flow matching learns a time dependent velocity field $v_\theta(x, \tau)$ whose flow carries a prior p_0 to the data p_1 . Instead of the intractable marginal velocity, one picks a conditional path with a known velocity u_τ and regresses onto it:

$$\mathcal{L}(\theta) = \mathbb{E}_{\tau, x_0, x_1} \left\| v_\theta(x_\tau, \tau) - u_\tau(x_\tau \mid x_0, x_1) \right\|^2. \quad (1)$$

The minimiser of (1) is the conditional expectation

$$v^*(x, \tau) = \mathbb{E}[u_\tau \mid x_\tau = x], \quad (2)$$

which is the marginal velocity whose flow maps p_0 to p_1 . The ablation of Section 5.1 uses (2) to compare the paths at their target.

The straight line path. The Euclidean path and its velocity are

$$x_\tau = (1 - \tau)x_0 + \tau x_1, \quad x_0 \sim \mathcal{N}(0, I), \quad u_\tau = x_1 - x_0. \quad (3)$$

The velocity is constant in τ , so the field stays close to straight, the property rectified and optimal transport flows are valued for [5, 6].

The geodesic path on the sphere. With the exponential and logarithm maps of the unit sphere, for unit vectors at angle $\theta = \arccos\langle \hat{x}_0, \hat{x}_1 \rangle$,

$$\text{Log}_{\hat{x}_0}(\hat{x}_1) = \theta \frac{\hat{x}_1 - \cos \theta \hat{x}_0}{\sin \theta}, \quad \text{Exp}_x(v) = \cos \|v\| x + \sin \|v\| \frac{v}{\|v\|}, \quad (4)$$

the geodesic $x_\tau = \text{Exp}_{x_0}(\tau \text{Log}_{x_0}(x_1))$ becomes spherical linear interpolation,

$$x_\tau = \frac{\sin((1 - \tau)\theta)}{\sin \theta} \hat{x}_0 + \frac{\sin(\tau\theta)}{\sin \theta} \hat{x}_1, \quad (5)$$

whose derivative in τ is the conditional velocity

$$u_\tau = \frac{\theta}{\sin \theta} \left[-\cos((1 - \tau)\theta) \hat{x}_0 + \cos(\tau\theta) \hat{x}_1 \right]. \quad (6)$$

Since $\|x_\tau\| = 1$, $\langle u_\tau, x_\tau \rangle = 0$, so the target velocity is tangent to the sphere. Equation (6) was checked both by hand and by automatic differentiation of (5). At sampling time the sphere arm projects its output onto the tangent space by $v \mapsto v - \langle v, x \rangle x$ and steps with the exponential map, which keeps every point on the sphere. The ninety degree spread in Figure 1 follows because independent uniform directions in $\mathbb{R}^D$ have inner product with mean zero and variance $1/D$.